

LECTURE 3: MONOPOLIES WITH CALCULUS

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WEEK 1

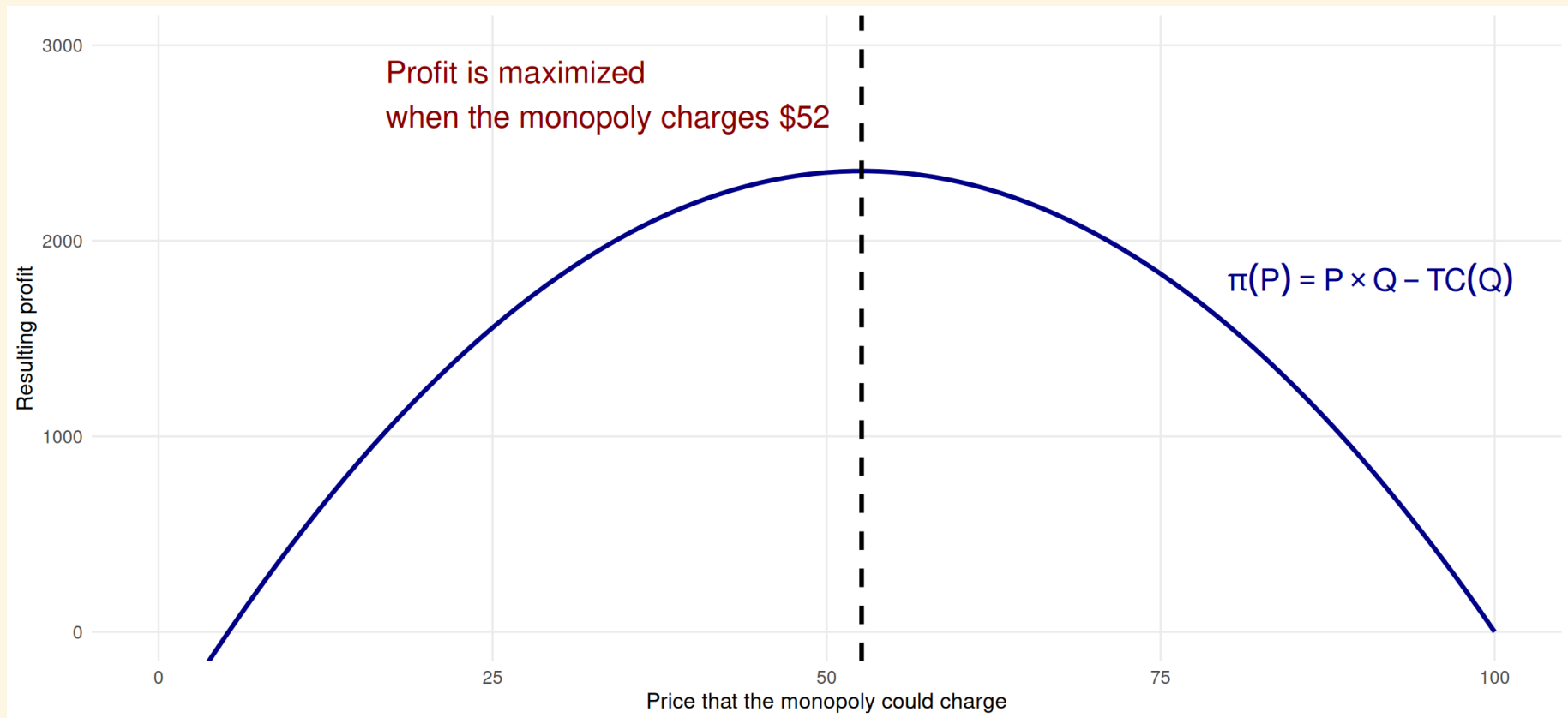
THE GOAL OF THE MONOPOLY IS TO MAXIMIZE PROFIT

The monopoly has the power to charge whatever price it likes

The smart monopoly will choose the price that maximizes its profit

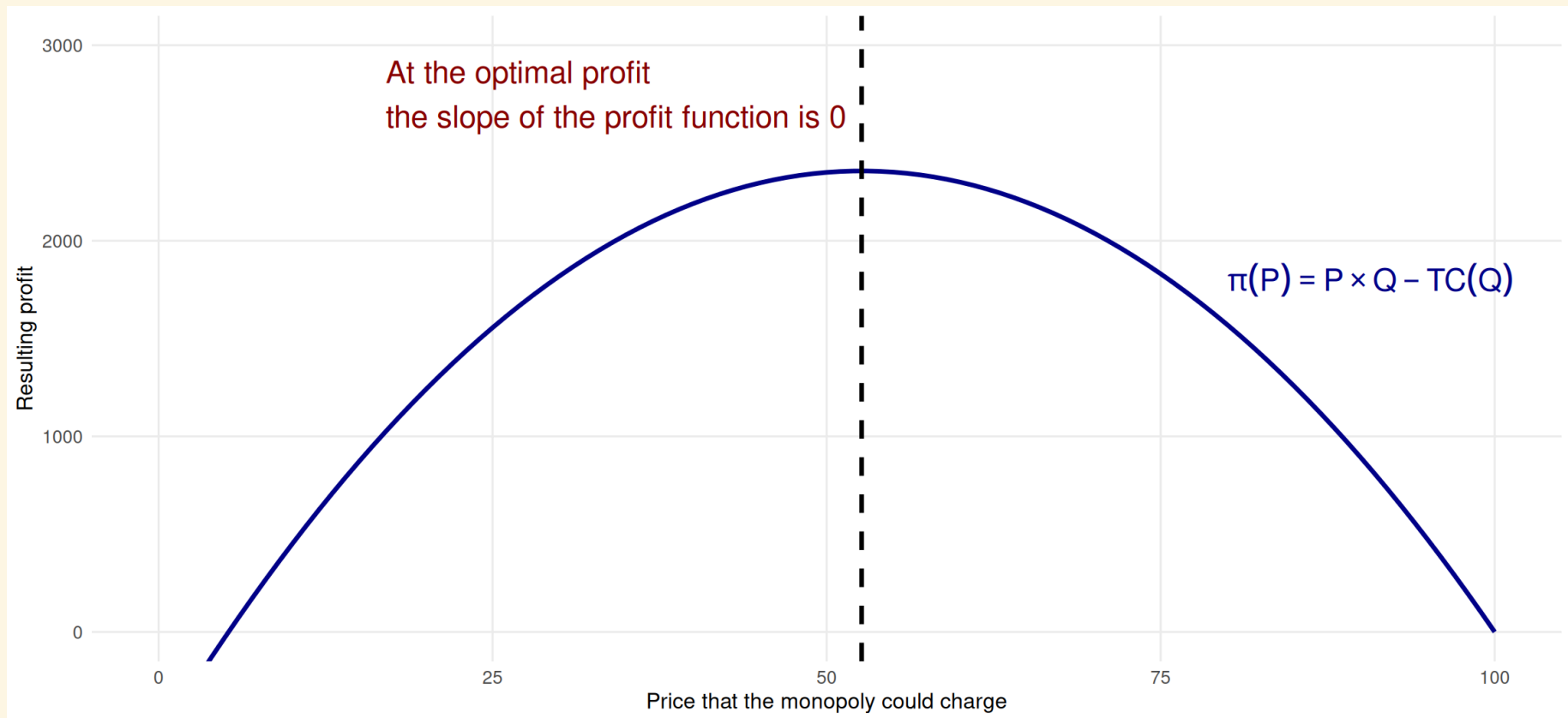
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So, if we know the profit function, we need to find the price so that:

$$\frac{\partial \pi}{\partial P} = 0$$

That is, the derivative of the profit function is zero when profit is maximized

Another way, if you drew the profit function for all possible prices the monopoly could charge, we're finding the price that gives us the highest profit

THE MONOPOLY'S TRADE-OFF IN CALCULUS

Last lecture we discussed that Nvidia's monopoly trade off from raising prices is that:

- Means it will sell fewer chips - losing sales but decreasing costs
- But it will make more profit on the chips that it does sell

Let's see how we can see that in calculus

THE MONOPOLY'S TRADE-OFF IN CALCULUS

We have that profit function is:

$$\pi = P \times Q(P) - TC(Q(P))$$

By the chain and product rules we get:

$$\frac{\partial \pi}{\partial P} = Q(P) + P \frac{dQ(P)}{dP} - \frac{dTC}{dQ} \frac{dQ(P)}{dP} = 0$$

We should think of the derivative as how much additional profit per change in price by a small, little bit

THE MONOPOLY'S TRADE-OFF IN CALCULUS:

$$\frac{\partial \pi}{\partial P} = Q(P) + P \frac{dQ(P)}{dP} - \frac{dTC}{dQ} \frac{dQ(P)}{dP} = 0$$

$Q(P)$ is how much additional revenue we make on the customers we keep by raising our prices

Remember, this is a change in profit per unit of price change. An approximation to our derivative is, if we raise our prices by \$x, we get:

$$\text{Change in revenue per \$ price change} = \frac{\text{Change in revenue}}{\text{Change in price}} = \frac{x \times \text{Sales quantity}}{x} = \text{Sales quantity}$$

THE MONOPOLY'S TRADE-OFF IN CALCULUS:

$$\frac{\partial \pi}{\partial P} = Q(P) + P \frac{dQ(P)}{dP} - \frac{dTC}{dQ} \frac{dQ(P)}{dP} = 0$$

$\frac{dQ(P)}{dP}$ is how many customers we lose from raising our prices. This will always be negative

So $P \frac{dQ(P)}{dP}$ is much revenue we lose from raising our prices because we drive customers out of the market

THE MONOPOLY'S TRADE-OFF IN CALCULUS:

$$\frac{\partial \pi}{\partial P} = Q(P) + P \frac{dQ(P)}{dP} - \frac{dTC}{dQ} \frac{dQ(P)}{dP} = 0$$

Together $Q(P) + P \frac{dQ(P)}{dP}$ is the change in revenue from raising the prices

We also call this marginal revenue - how much revenue changes from a small change in prices: $MR(P) = Q(P) + P \frac{dQ(P)}{dP}$

THE MONOPOLY'S TRADE-OFF IN CALCULUS:

$$\frac{\partial \pi}{\partial P} = Q(P) + P \frac{dQ(P)}{dP} - \frac{dTC}{dQ} \frac{dQ(P)}{dP} = 0$$

Lastly, our costs also change as we sell fewer goods. We call this our marginal cost:

$$MC(P) = \frac{dTC}{dQ} \frac{dQ(P)}{dP}$$

THE MONOPOLY'S TRADE-OFF IN CALCULUS:

$$\frac{\partial \pi}{\partial P} = \underbrace{Q(P) + P \frac{dQ(P)}{dP}}_{\text{Marginal revenue}} - \underbrace{\frac{dTC}{dQ} \frac{dQ(P)}{dP}}_{\text{Marginal cost}} = 0$$
$$Q(P) + P \frac{dQ(P)}{dP} = \frac{dTC}{dQ} \frac{dQ(P)}{dP}$$
$$MR(P) = MC(P)$$

So our trade-off, in calculus terms, means we're going to keep raising prices until the change in revenue (marginal revenue) equals the change in costs (marginal cost)

EXAMPLE TO DO ON THE BOARD

Nvidia has a constant marginal cost of making a chip of \$2,000

Demand for chips is $Q_D = 10,000 - 1.5P$

Let's solve for how much Nvidia would charge for chips

EXAMPLE TO DO ON THE BOARD

First, we find Nvidia's profit function - this is how much profit it would make for some given P .

We don't know what the optimal P is yet - we want to solve for it

The profit function is:

$$\pi = \underbrace{P \times (10,000 - 1.5P)}_{\text{Revenue} = \text{Price times demand}} - \underbrace{2,000 \times (10,000 - 1.5P)}_{\text{Total cost}}$$

Because marginal cost is constant, total cost is Marginal cost $\times$ Quantity demanded

EXAMPLE TO DO ON THE BOARD

We'll expand out the brackets of the profit function

$$\begin{aligned}\pi &= 10,000P - 1.5P^2 - 20,000,000 + 3,000P \\ &= 13,000P - 1.5P^2 - 20,000,000\end{aligned}$$

Then we want to find the price where profit is maximized. This means we find the derivative of profit, and solve for the price that makes it equal to 0

This corresponds to the top of the parabola in the example diagram:

$$\frac{\partial \pi}{\partial P} = 13,000 - 3P \stackrel{\text{set}}{=} 0$$

We then solve for P:

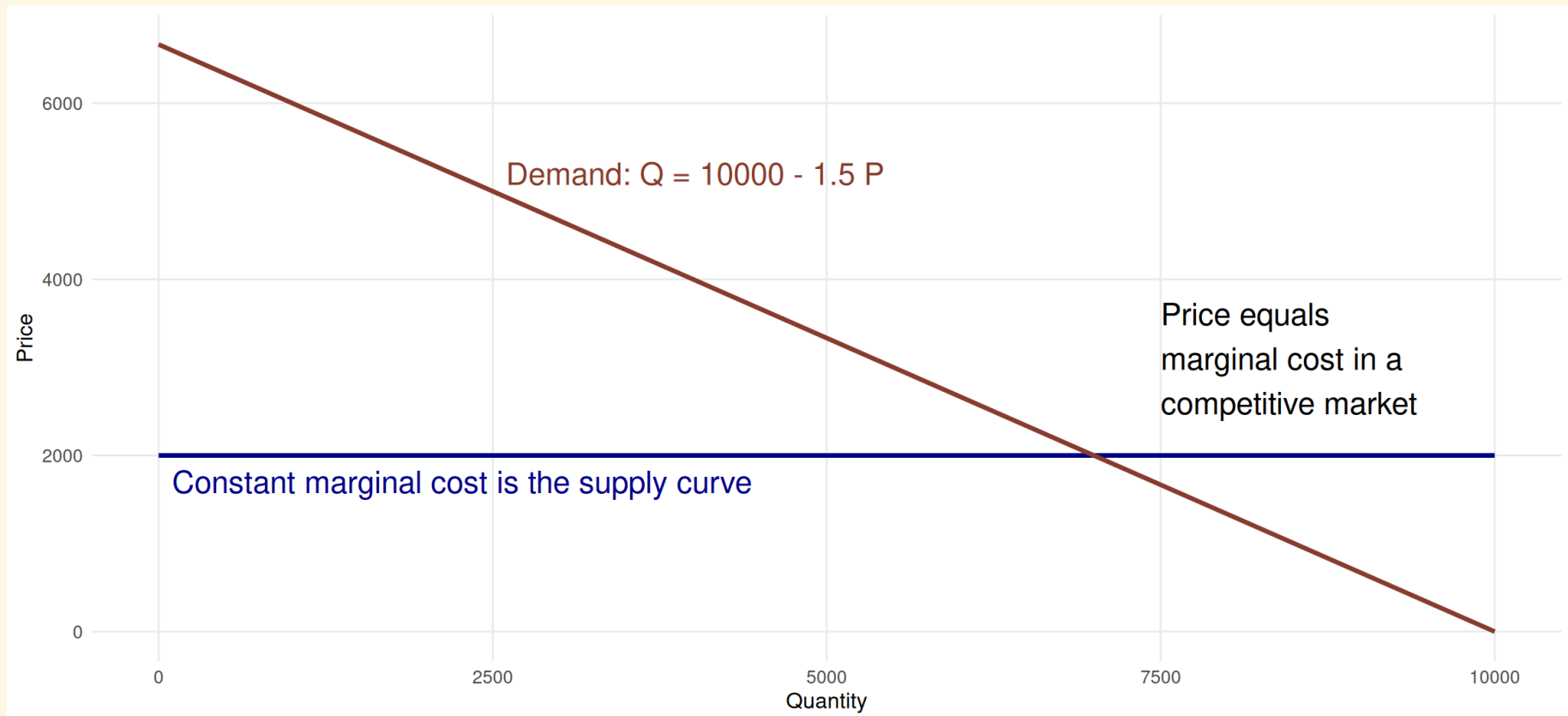
$$\begin{aligned}13,000 - 3P &\stackrel{\text{set}}{=} 0 \\ 13,000 &= 3P \\ P &= \$4,333\end{aligned}$$

A \$2,333 mark up over marginal cost!

And we would sell $Q_D = 10,000 - 1.5 \times 4,333 = 3,500$ chips

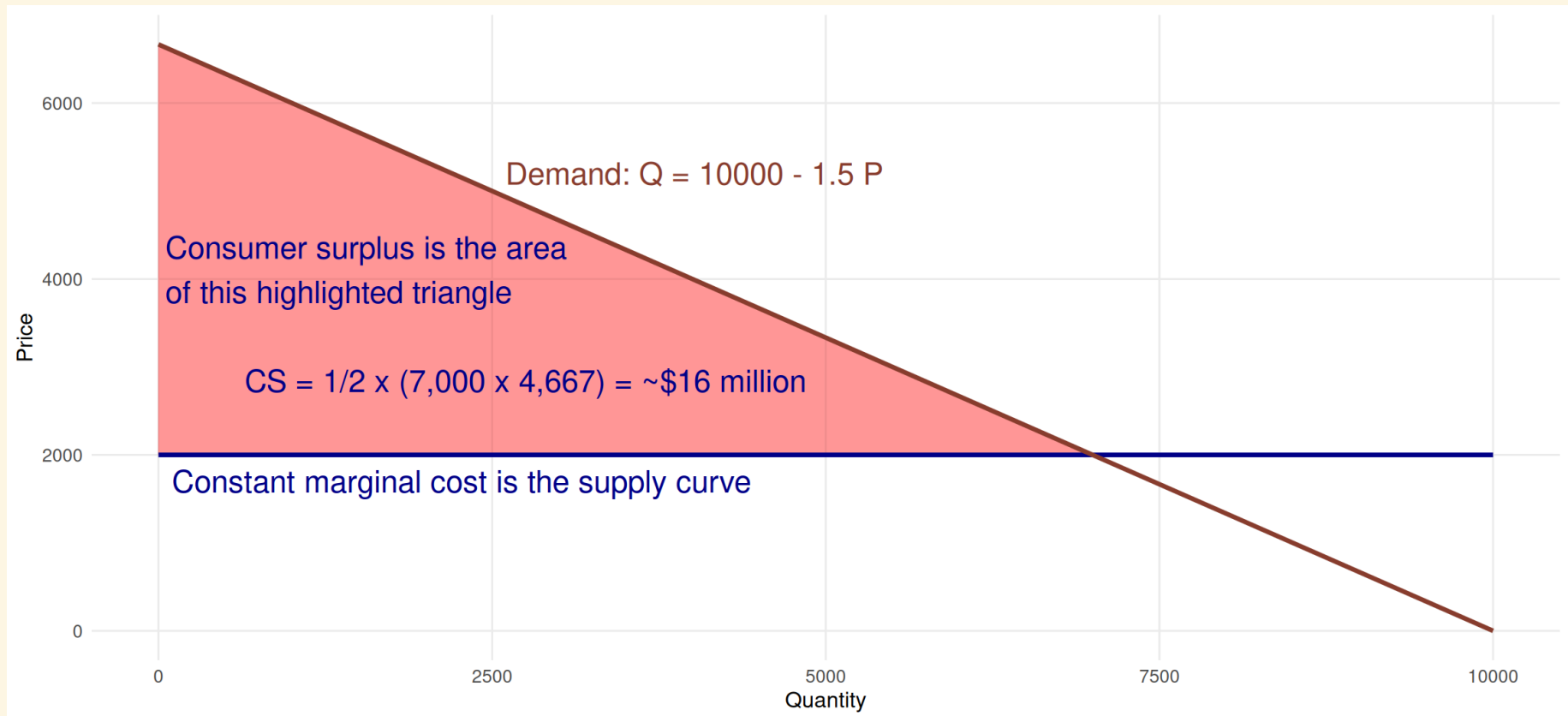
WELFARE IMPLICATIONS OF A MONOPOLY

When thinking of welfare implications, we often compare to a competitive market - where Nvidia prices at marginal cost = \$2,000



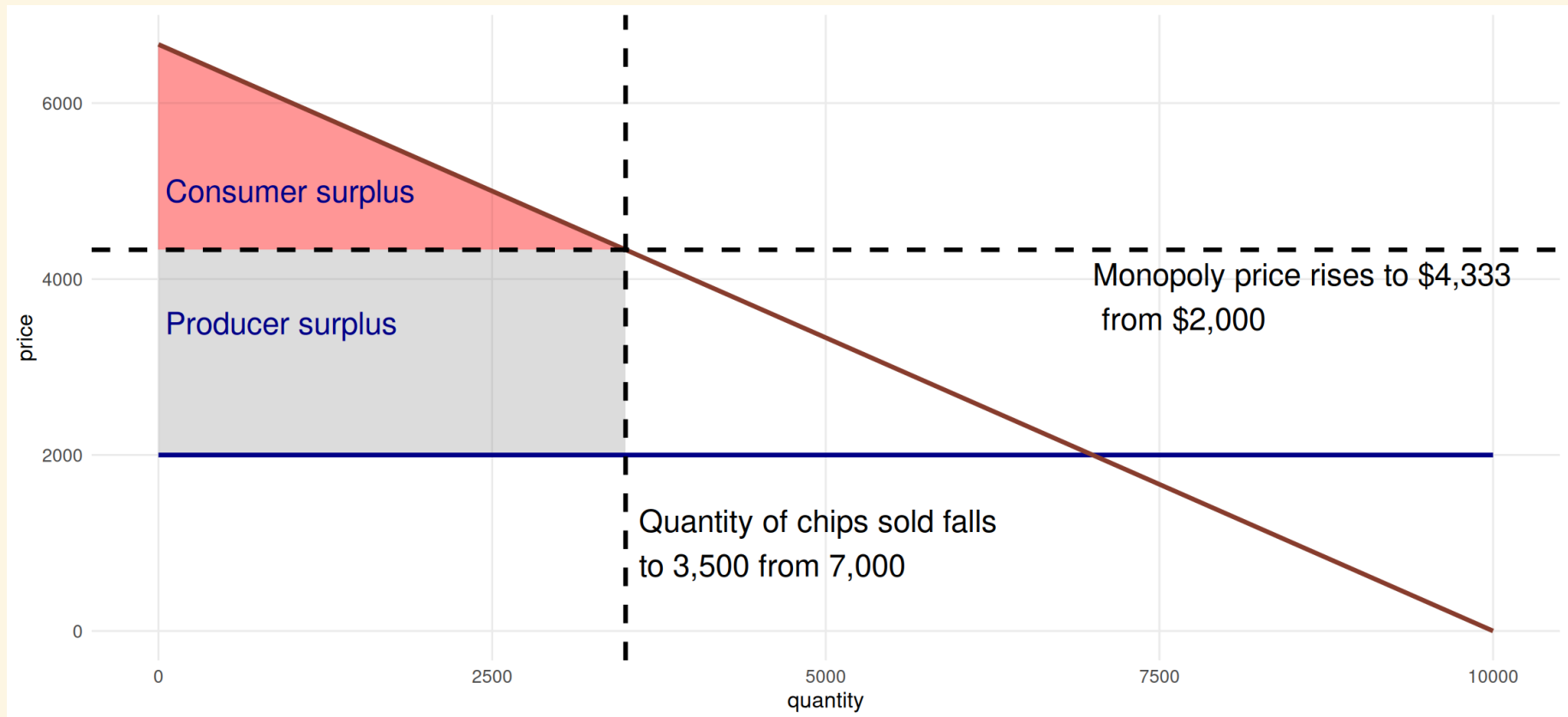
WELFARE IMPLICATIONS OF A MONOPOLY

The consumer surplus in a competitive market is the area between the demand curve and price



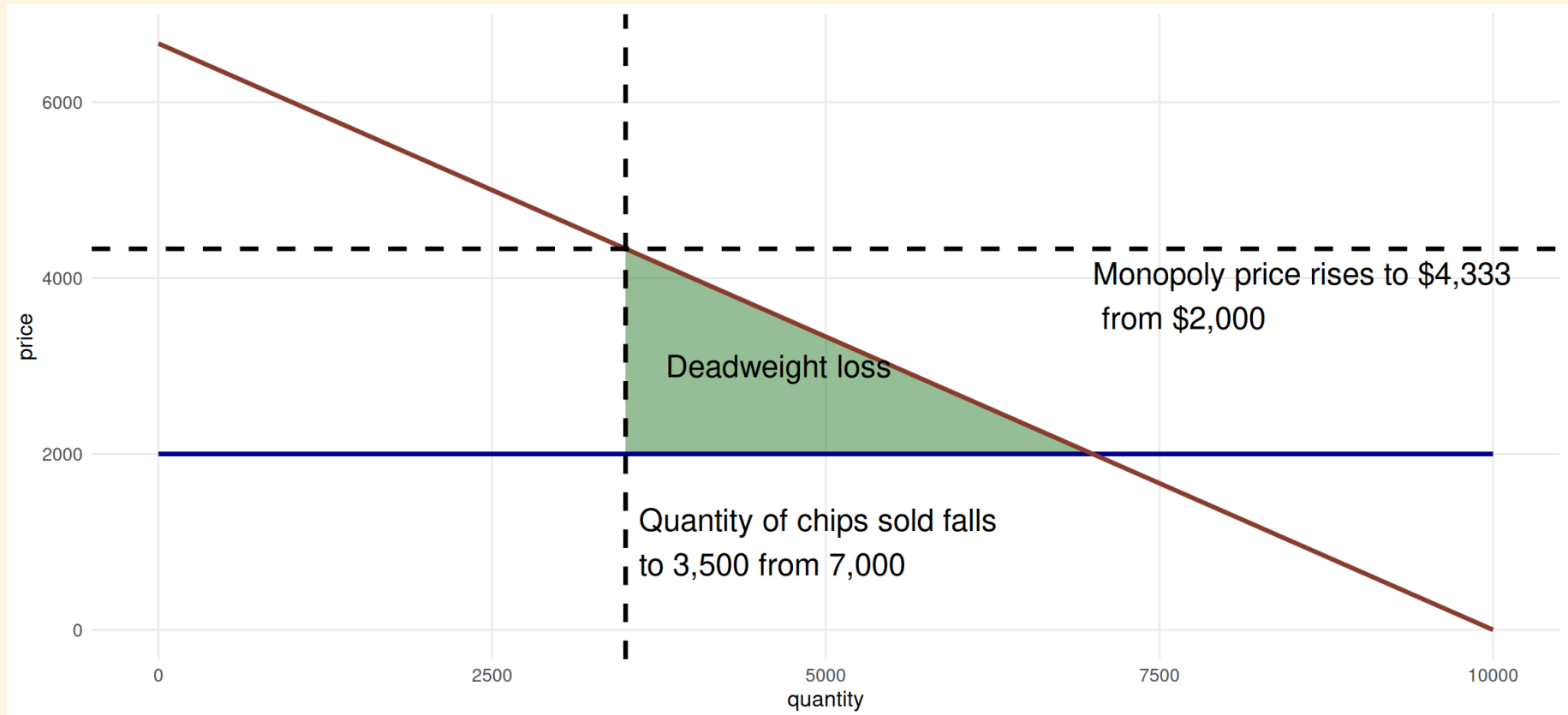
WELFARE IMPLICATIONS OF A MONOPOLY

Let's calculate consumer and producer surplus under a monopoly on the board



WELFARE IMPLICATIONS OF A MONOPOLY

This triangle is the deadweight loss - how much worse off society is from a monopoly compared to a competitive market



WELFARE IMPLICATIONS OF A MONOPOLY

This triangle is the deadweight loss - how much worse off society is from a monopoly compared to a competitive market

It is the lost surplus of customers that have willingness to pay above the marginal cost

These people would be willing to buy Nvidia chips for a profitable price! But the monopoly shuts them out of the market

This is why we typically don't like market power - it prevents mutually beneficial activity from occurring

